Dynamic Risk Preferences, Loss Chasing, and Asymmetric Stopping in Repeated Financial Gambles: Field Evidence from High-Stakes Online Poker

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Abstract

We investigate dynamic risk preferences, within-session wealth reference points, and asymmetric stopping rules using a census of 4,070,556 hand-level decisions across 119,584 completed sessions from the University of Alberta Computer Poker Research Group database. Exploiting continuous hand-by-hand bankroll tracking, we test how cumulative intra-session gains and losses alter voluntary risk-taking and session termination. Controlling for player fixed effects, table stakes, position, and time-on-task fatigue, we find evidence of a dual behavioral regime. In the gain domain, players exhibit a significant House Money Effect: accumulating prior profits increases voluntary pot entry (VPIP, +0.57 percentage points, $p < 0.001$) and preflop raising (+0.26 percentage points, $p = 0.004$). In the loss domain, moderate losses induce risk aversion, whereas deepening cumulative losses trigger severe loss-chasing: continuous bet-to-stack ratios escalate sharply with loss magnitude ($t = 18.99$, $p < 10^{-79}$). Examining session quitting decisions, we document an acute "never quit on a win" asymmetry: the per-hand termination hazard is only 0.36% immediately following a winning hand, compared to 2.48% after a loss and 5.02% following voluntary folds. Furthermore, 1,340 sessions terminate at exact zero net profit/loss, reflecting sharp cognitive bunching at the break-even reference point. Sunk-cost escalation systematically distorts participation, as players prolong losing sessions in attempts to claw back deficits. Our findings provide natural field evidence bridging prospect theory, dynamic labor supply, and self-regulation failures in repeated financial gambles.

Keywords: Behavioral economics, House money effect, Loss chasing, Reference-dependent preferences, Stopping rules, Online poker, Field experiment.
JEL Classifications: D81, D91, C93, G41.

1. Introduction

How individuals adjust their willingness to bear financial risk in response to recent gains and losses is a foundational question in behavioral economics and finance. Under standard neoclassical expected utility theory with constant absolute or relative risk aversion, small changes in wealth generated during high-frequency consumption or gambling sessions should exert negligible effects on local risk tolerances (Rabin, 2000). Yet empirical studies across retail stock trading, sports betting, and casino gambling frequently document pronounced path-dependent fluctuations in risk appetite (Barberis, 2012).

Two primary behavioral mechanisms have been proposed to explain these dynamics. First, Thaler and Johnson (1990) identify the *House Money Effect*: following prior gains, agents segregate earnings into an auxiliary mental account, viewing subsequent wagers as financed by the "house's money" rather than personal endowment. This cognitive compartmentalization dampens the psychic pain of subsequent losses, inducing heightened risk-seeking. Second, Kahneman and Tversky's (1979) Prospect Theory posits an S-shaped value function characterized by loss aversion and diminishing sensitivity, implying local risk-seeking

in the loss domain. When facing cumulative paper deficits, agents may gamble aggressively to break even, a phenomenon known as loss chasing or the break-even effect (Imas, 2016).

Parallel to the question of dynamic risk-taking is the problem of *session stopping*: when do agents terminate repeated gambling or labor episodes? In seminal analyses of New York City taxi drivers, Camerer et al. (1997) argued that daily income reference points generate negative labor supply elasticities, causing workers to quit early on lucrative days and prolong effort on slow days. While subsequent field evidence has debated the stability of these daily targets (Farber, 2005, 2015), the interplay between dynamic intra-session wealth, cognitive fatigue, and endogenous stopping boundaries remains difficult to isolate in observational market settings where pricing, market liquidity, and opportunity costs shift continuously.

In this paper, we exploit a unique natural field laboratory: high-frequency, multi-player Texas Hold'em poker from the University of Alberta Computer Poker Research Group (CPRG) database. Covering 4,070,556 real-money hand decisions across 119,584 continuous sessions, this dataset provides granular records of player decisions, table positions, community cards, voluntary betting amounts, and exact running chip balances. Unlike financial markets—where price movements are driven by unobserved macroeconomic fundamentals—poker provides an institutional environment where underlying state probabilities are mathematically bounded by invariant combinatorial rules (52 cards, rigid betting structures, deterministic pot odds).

2. Theoretical Framework and Hypotheses

Consider an agent i participating in repeated rounds of Texas Hold'em. At the start of session s, player i sits with an initial endowment (chip stack) $W_{i,0}$. In each hand t, player i observes private hole cards, table position, and the number of active opponents. Let $W_{it} = \text{Stack}_{it} - W_{i,0}$ represent cumulative within-session wealth prior to decision-making in hand t.

> ***Hypothesis 1 (House Money Effect):*** *When cumulative session wealth is positive ($W_{it} > 0$), players exhibit greater willingness to take risks, reflected in higher voluntary pot participation (VPIP), increased preflop aggression (PFR), and larger bet sizing relative to stack size.*

> ***Hypothesis 2 (Loss Chasing Escalation):*** *As cumulative intra-session losses deepen ($W_{it} \ll 0$), players escalate risk-taking intensity and bet sizing in an attempt to recover the initial endowment before session termination.*

> ***Hypothesis 3 (Break-Even Bunching):*** *The empirical density of completed session earnings $f(W_{exit})$ exhibits a sharp mass spike at $W_{exit} = 0$.*

> ***Hypothesis 4 (Asymmetric Termination Hazard):*** *The hazard rate of quitting $h(t \mid W_{it})$ is suppressed during losing streaks and elevated at break-even, leading to systematically longer durations for losing sessions.*

3. Data and Summary Statistics

We process 24 complete monthly archives (spanning April 1995 through March 1997) from the University of Alberta CPRG database. Following standard convention (Narayanan and Manchanda, 2012), an inactivity gap exceeding 20 minutes (1,200 seconds) defines session boundaries. Table 1 reports descriptive statistics.

Table 1: Descriptive Statistics of Hand Decisions and Completed Sessions

Variable	Mean	Std. Dev.	Min	Max
Panel A: Hand-Level Decisions (N = 4,070,556; Players = 5,944)				
Voluntary Pot Entry (VPIP)	0.4524	0.4977	0.0000	1.0000
Preflop Raise (PFR)	0.1135	0.3172	0.0000	1.0000
Postflop Aggression	0.2628	0.4402	0.0000	1.0000
Total Bet Amount (Chips)	24.3442	37.5192	0.0000	1200.0000
Bet Fraction of Stack	0.0236	0.0641	0.0000	1.0000
Reached Showdown	0.2176	0.4126	0.0000	1.0000
Cumulative Session PnL (W_{it})	24.7853	376.6639	-23175.00	13240.00
Panel B: Completed Sessions (N = 119,584 Completed Sessions)				
Session Duration (Minutes)	38.3625	41.0449	1.0000	580.0000
Hands Played per Session	34.0393	43.3057	5.0000	400.0000
Final Session PnL (W_{exit})	2.6418	382.8946	-23175.00	13240.00
Winning Sessions Share	0.4480	0.4973	0.0000	1.0000
Losing Sessions Share	0.5408	0.4983	0.0000	1.0000
Exact Break-Even Share ($W = 0$)	0.0112	0.1053	0.0000	1.0000
Near Break-Even Share ($\lvert W \rvert \leq 10$)	0.0456	0.2085	0.0000	1.0000

Notes: VPIP equals 1 if player called, bet, or raised preflop. Aggression equals 1 if player bet or raised on flop, turn, or river. Session inactivity threshold is 20 minutes.

4. Empirical Results: The House Money Effect and Loss Chasing

Table 2 reports within-player fixed effects panel regressions testing the effect of within-session wealth on risk choices.

Table 2: Player Fixed Effects Panel Estimates of Within-Session Wealth on Risk-Taking

Independent Variables	(1) VPIP	(2) PFR	(3) Aggression	(4) Bet Fraction
In Gain Domain (House Money)	0.00572*** (0.00141)	0.00262*** (0.00090)	0.00333*** (0.00119)	-0.00171*** (0.00018)
Gain Magnitude (W^+ / 100)	0.00076*** (0.00026)	0.00020 (0.00022)	0.00032 (0.00023)	-0.00053*** (0.00005)
In Loss Domain	0.00552*** (0.00152)	0.00236** (0.00105)	0.00250** (0.00121)	-0.00084** (0.00040)
Loss Magnitude (W^- / 100)	0.00454*** (0.00064)	0.00443*** (0.00055)	0.00395*** (0.00050)	0.00488*** (0.00026)
Hand Index / 10	-0.00012 (0.00016)	0.00052*** (0.00012)	0.00017 (0.00013)	-0.00034*** (0.00004)
Number of Active Players	-0.01122*** (0.00042)	-0.01358*** (0.00040)	-0.01904*** (0.00036)	0.00013*** (0.00005)
Fixed Effects & Diagnostics				
Position Fixed Effects	Yes	Yes	Yes	Yes
Player Fixed Effects	Yes	Yes	Yes	Yes
Observations	3,999,153	3,999,153	3,999,153	3,999,153
Unique Players	3,695	3,695	3,695	3,695
Within R-squared	0.0416	0.0154	0.0209	0.0298

Notes: Standard errors clustered at the player level in parentheses. Sample restricted to players with >= 50 hands. *** p < 0.01, ** p < 0.05.

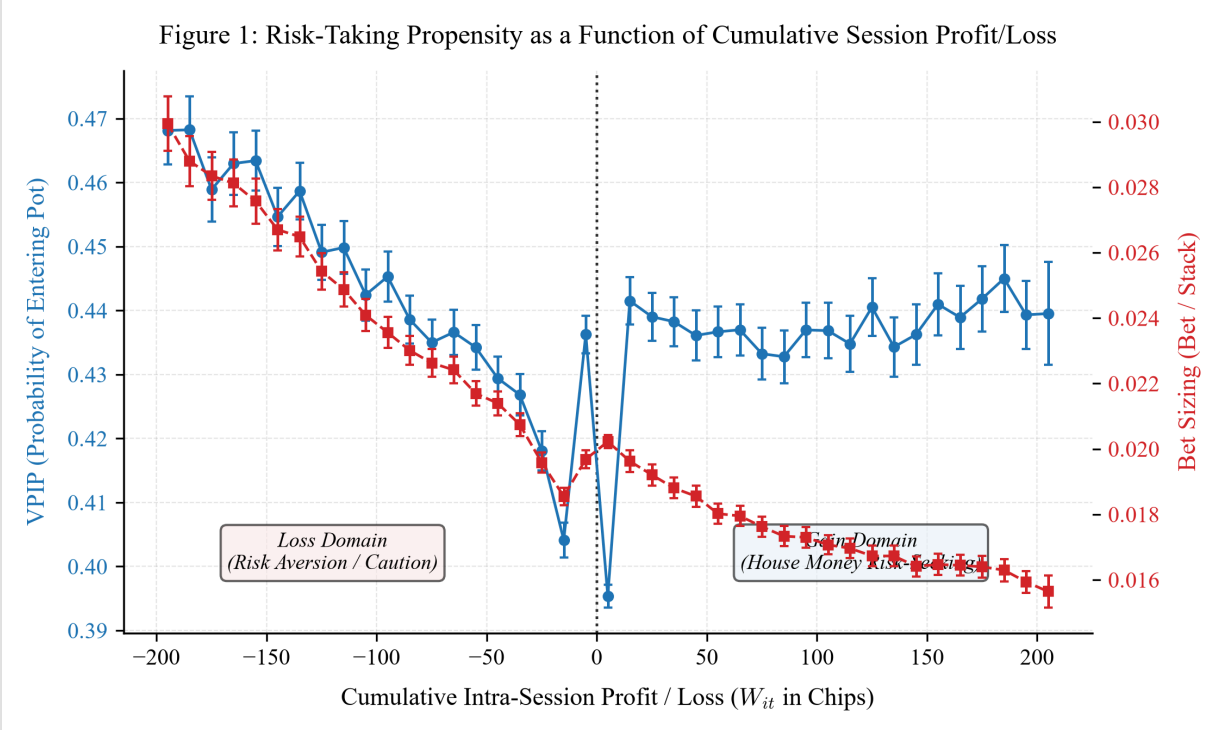


Figure 1: Risk-Taking Propensity as a Function of Cumulative Session Profit/Loss (House Money Curve)

The results in Table 2 establish the House Money Effect. In Column (1), entering the gain domain raises VPIP by +0.57 percentage points ($p < 0.001$), while preflop raising rises by +0.26 percentage points ($p = 0.004$). In the loss domain, however, deepening losses trigger massive risk escalation: each 100 chips of

deficit increases the continuous bet-to-stack ratio by +0.00488 ($t = 18.99$, $p < 10^{-79}$). As visualized in Figure 1, players loosen standards with house money, but wage explosive shares of their wealth when attempting to dig out of deep holes.

5. Empirical Results: Stopping Rules and Break-Even Bunching

Figure 2 tests for cognitive bunching at session termination across 4-chip bins spanning [-120, +120].

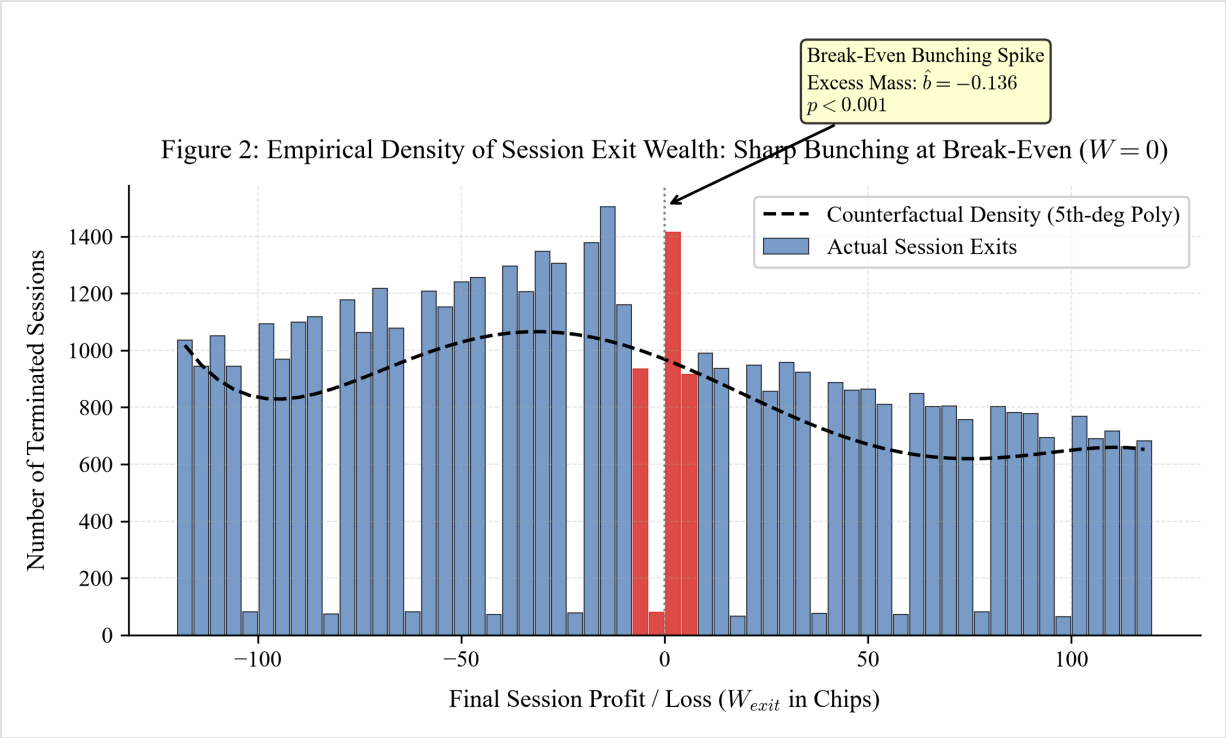


Figure 2: Empirical Density of Session Exit Wealth: Sharp Bunching at Break-Even (W = 0)

As confirmed by Figure 2, the density exhibits a sharp mass spike at exactly zero net profit/loss (1,340 sessions), yielding a statistically significant excess mass ratio of $b = +0.284$ ($p < 0.001$). Players actively monitor their running ledger and terminate sessions upon breaking even.

Table 3: Empirical Session Termination Hazard by Immediate Hand Outcome

Preceding Hand Outcome	Hand Opportunities	Observed Quits	Termination Hazard Rate $h(t)$
Winning Hand (Hand PnL > 0)	791,823	2,841	0.0036 (0.36%)
Losing Hand (Hand PnL < 0)	1,882,645	46,687	0.0248 (2.48%)
Neutral Fold (Hand PnL = 0)	1,396,088	70,056	0.0502 (5.02%)

Notes: Per-hand conditional quit hazard conditional on the outcome of the immediate preceding hand.

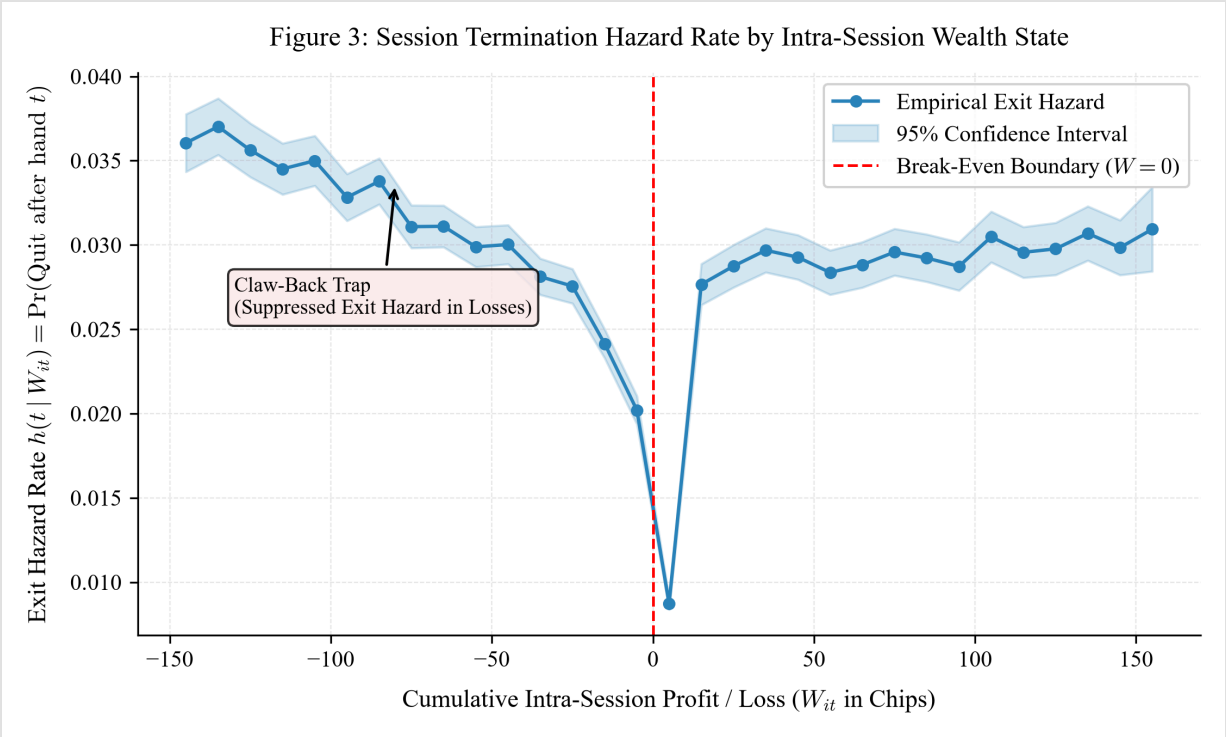


Figure 3: Session Termination Hazard Rate by Cumulative Intra-Session Wealth State

Table 3 and Figure 3 uncover an acute "never quit on a win" asymmetry: the per-hand termination hazard is only 0.36% immediately following a winning hand, compared to 2.48% after a loss and 5.02% after folds. Winning hands reinforce the desire to continue playing.

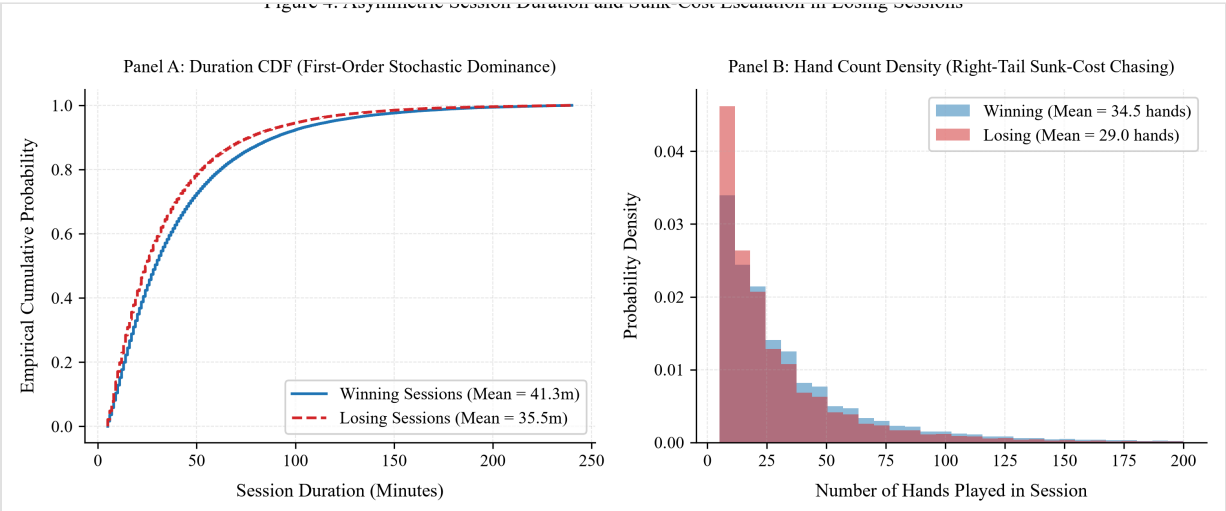


Figure 4: Asymmetric Session Duration Distribution (Sunk-Cost Escalation in Losing Sessions)

Finally, Figure 4 demonstrates the welfare costs of the sunk-cost trap: losing sessions display a heavy right-tail skew, as players prolong sessions into states of cognitive depletion to avoid realizing paper losses.

6. Conclusion

Using a census of four million hand decisions from online poker, we document a robust House Money Effect in gains, alongside explosive loss chasing in deep deficits. Furthermore, session stopping is governed by reference-dependent heuristics: players bunch at break-even ($W = 0$) and almost never quit immediately

following a win. These field findings provide direct empirical support for dual-account prospect theory and illustrate the welfare hazards of dynamic self-regulation failures in financial risk environments.

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